
SIMULATING BIOLOGICAL INTELLIGENCE: ACTIVE INFERENCE WITH EXPERIMENT-INFORMED GENERATIVE MODEL

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ABSTRACT

With recent and rapid advancements in artificial intelligence (AI), understanding the foundation of purposeful behaviour in autonomous agents is crucial for developing safe and efficient systems. While artificial neural networks have dominated the path to AI, recent studies are exploring the potential of biologically based systems, such as networks of living biological neuronal networks. Along with promises of high power and data efficiency, these systems may also inform more explainable and biologically plausible models. In this work, we propose a framework rooted in active inference, a general theory of behaviour, to model decision-making in embodied agents. Using experiment-informed generative models, we simulate decision-making processes in a simulated game-play environment, mirroring experimental setups that use biological neurons. Our results demonstrate learning in these agents, providing insights into the role of memory-based learning and predictive planning in intelligent decision-making. This work contributes to the growing field of explainable AI by offering a biologically grounded and scalable approach to understanding purposeful behaviour in agents.

Keywords Generative Models · Active Inference · Counterfactual Learning · Biologically Inspired AI · Explainable Artificial Intelligence

1 Introduction

The rapid advancement of artificial intelligence (AI) underscores a critical need for a more robust foundation for purposeful decision-making in autonomous agents, essential for developing safe and efficient systems. While artificial neural networks (ANNs) have paved many paths in AI, their limitations in emulating the nuanced, adaptive behaviours observed in biological systems are becoming increasingly apparent. This recognition has spurred exploration into biological paradigms, where living neuronal networks (BNNs) may offer profound insights and efficiencies currently beyond the reach of conventional ANNs.

Recent experimental work, such as the DishBrain system Kagan et al. [2022], exemplifies this potential. In these experiments, cortical neurons cultured on silicon chips and integrated into a closed-loop simulated game environment demonstrated the ability to modify their behaviour in real time. This finding, where even minimal biological substrates exhibit adaptive intelligence through sensory feedback Kagan et al. [2022], has helped establish synthetic biological intelligence (SBI), a paradigm leveraging living neural systems as computational substrates Kagan et al. [2023]. Such biologically grounded approaches, alongside novel explorations in natural computation like DNA-based molecular learning systems Evans et al. [2024], are expanding the horizons of AI by emphasising biological plausibility.

Despite these significant experimental strides, a crucial gap persists: the lack of theoretical frameworks capable of comprehensively modelling and explaining the emergent intelligence observed in these biological systems. Existing AI models, particularly those based on deep learning, often fall short in capturing the adaptive, memory-driven, and

embodied nature inherent in biological decision-making. To address this, our work introduces a generative model inspired by the DishBrain experiments (see Section 2.2), leveraging active inference as a theoretical framework for modelling behaviour. This approach allows us to simulate decision-making processes, focusing on how agents learn and engage in predictive planning within a simulated game-play environment mirroring biological experiments (as detailed in Section 3.1).

Our simulations demonstrate how these agents learn, offering insights into the interplay of memory-based learning and predictive planning (Section 3.3). Furthermore, we systematically compare various decision-making schemes (Section 3.2) and discuss the broader implications of our findings for neuroscience, AI, and embodied intelligence in Section 4. Ultimately, this study aims to advance the development of explainable, biologically inspired AI by integrating biologically plausible mechanisms with robust computational modelling. Our results highlight the pivotal role of memory in decision-making, offering a contrast to conventional planning-centric AI, and contribute to the creation of more interpretable and adaptive intelligent systems.

2 Methods

2.1 Generative models based on POMDPs

A generative model is a scaled-down version of the natural world, but not too simple Friston et al. [2023a]. The purpose is to enable the agent to have prior expectations (about incoming observations) and simultaneously predict future observations to control them in its favour. So, active inference, at its core, is a model-based framework. However, it does not impose a particular structure on such models. Several models are viable for these problems, including Partially Observable Markov Decision Processes (POMDPs) Kaelbling et al. [1998], Lovejoy [1991] and Gaussian mixture models Pernkopf and Bouchaffra [2005], depending on the specific requirements of the task. Generative models are gaining popularity in machine learning for various reasons, like the emergence of ChatGPT Kocón et al. [2023]. Their popularity is also emerging in fields including weather modelling Jin et al. [2023] and protein engineering Ingraham et al. [2023]. In this paper, we stick to the POMDP architecture, assuming the agent encodes a discrete world model Da Costa et al. [2020] (See Sec.2.1).

POMDPs offer a universal structure to model discrete state-space environments where parameters can be expressed as tractable categorical distributions [Kaelbling et al., 1998]. A POMDP can be formally defined as a tuple of finite sets $(S, O, U, \mathbb{B}, \mathbb{A})$:

- $s \in S$: S is a set of hidden states (s) causing observations o .
- $o \in O$: O is a set of observations, where $o = s$, in the fully observable setting. In a partially observable setting, $o = f(s)$.
- $u \in U$: U is a set of actions (u).
- $\mathbb{B}$: encodes the one-step transition dynamics, $P(s_t|s_{t-1}, u_{t-1})$ i.e., the probability of transitioning to state s_t at time t given that action u_{t-1} is taken in state s_{t-1} at time $t - 1$.
- $\mathbb{A}$: encodes the likelihood mapping, $P(o_\tau|s_\tau)$, for the partially observable setting.
- $\mathbb{D}$: Encodes the prior of the agent about the hidden state factor s .
- $\mathbb{E}$: Encodes the prior of the agent about actions u .

In a POMDP, the hidden states (s) generate observations (o) through the likelihood mapping ($\mathbb{A}$) in the form of a categorical distribution, $P(o_\tau|s_\tau) = \text{Cat}(\mathbb{A} \times s_\tau)$.

$\mathbb{B}$ is a collection of square matrices $\mathbb{B}_u$, where $\mathbb{B}_u$ represents transition dynamics $P(s_t|s_{t-1}, u_{t-1} = u)$: ($\mathbb{B}$) determines the dynamics of s given the agent’s action u as $P(s_t|s_{t-1}, u_{t-1}) = \text{Cat}(\mathbb{B}_{u_{t-1}} \times s_{t-1})$. In $[\mathbb{A} \times s_\tau]$ and $[\mathbb{B}_{u_\tau} \times s_\tau]$, s_τ is represented as a one-hot vector that is multiplied through regular matrix multiplication¹.

The *Markovianity* of POMDPs means that state transitions are independent of history (i.e. state s_t only depends upon the state-action pair (s_{t-1}, u_{t-1}) and not s_{t-2}, u_{t-2} etc.). The generative model can be summarised as follows,

$$P(o_{1:t}, s_{1:t}, u_{1:t}) = P(\mathbb{A})P(\mathbb{B})P(\mathbb{D})P(\mathbb{E}) \prod_{\tau=1}^t P(o_\tau|s_\tau, \mathbb{A}) \prod_{\tau=2}^t P(s_\tau|s_{\tau-1}, u_{\tau-1}, \mathbb{B}). \quad (1)$$

¹One-hot is a group of bits among which the legal combinations of values are only those with a single high (1) bit and all the others low (0). Here, the bit (1) is allocated to the state $s = s_\tau$

So, from the agent’s perspective, when encountering a stream of observations in time, such as $(o_1, o_2, o_3, \dots, o_t)$, as a consequence of performing a series of actions $(u_1, u_2, u_3, \dots, u_{t-1})$, the generative model quantitatively couples and quantifies the causal relationship from action to observation through some assumed hidden states of the environment. These are called ‘hidden’ states because the agent cannot directly observe the hidden states s in the environment. The agent maintains beliefs about s , using the observations o . In the experiment (Fig. A.1), the observations directly represent the state parameters, such as the ball position without noise. Hence, the problem is more control-oriented than a perception problem. So, throughout the paper, we assume $\mathbb{A}$: that encodes the likelihood mapping ($P(o_\tau | s_\tau)$) is an identity distribution.

Based on this representation, an agent can now attempt to optimise its actions to keep receiving preferred observations. So far, the introduced generative model has no concept of ‘preference’ and ‘goal’. In the next few sections, we focus on decision-making schemes in active inference, addressing variations of algorithms used to model ‘goal-directed’ (i.e. purposeful) behaviour. In the next section, we lay out the experiment-informed generative model we designed and used throughout the paper.

2.2 Experiment informed generative model

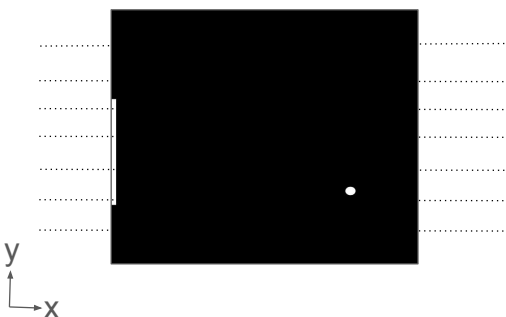
A. Structure: - State space ($38 * 8 * 8 = 2432$ states) - Action space (Up, Down, and Stay) - Observations (Fully observable states.)			E. Parameters of interest: - B (Transition dynamics in POMDP) - CL (Counterfactual state-action mapping) Γ (Risk, CL method) - C (Prior preference distribution, learned for DPEFE and AIF)
B. X-coordinate of ball (ball-x) (Assumed since ball-x is communicated to ‘DishBrain’ through 38 distinct frequencies (4-41 Hz) denoting evenly spaced positions, i.e 38 states.)	C. Y-coordinate of ball (ball-y) (Assumed since ball-y communicated through 8 stimulus electrodes which are evenly spaced along the y-axis generating 8 possible place-coded positions.)	D. Y-coordinate of paddle (paddle-y) (Assumed to be part of the generative model as control is exerted. Modality is similar to ball’s y-coordinate, i.e 8 states.)	

Figure 1: A: Layouts the basic structure of the generative model with dimensions of difference modalities. The corresponding dimensions (38,8, and 8) are justified in B, C, and D. B, C, D: The observation modalities involved are ‘ball-x’, ‘ball-y’, and ‘paddle-y’. The dimensions of each modality are determined according to the experiment specifications. E: The parameters of interest in the generative model used in each decision-making algorithm mentioned in this paper.

State-outcome modalities The information about the Pong game states was communicated to the population of neurons via the DishBrain system using biologically safe voltage and frequency levels in the experiment. We assume three state-observation modalities corresponding to the input signals in the generative model: ‘ball-x’, ‘ball-y’, and ‘paddle-y’. They represent the ball’s x and y coordinates and the y coordinate of the middle of the paddle, respectively. In the experiment, the x coordinate of the ball represents its distance from the paddle along the x -axis. The experiment communicates it through a rate-coded component of 38 distinct frequencies ranging from 4 – 41 Hertz. The y coordinate of the ball is transmitted through one of the 8 evenly distributed stimulation electrodes on the y -axis of HD-MEA (See Fig. 1 B, C and D). Using this information, we can set up the structure of the generative model.

Structure We assume the state space of the generative model has 2432 states (See Fig. 1, A) in correspondence with the dimensions mentioned above. 38 distinct states representing ‘ball-x’, and 8 different states for each of the ‘ball-y’ and ‘paddle-y’, respectively. The action space has three elements, namely ‘Up’, ‘Down’, and ‘Stay’, the same as in the simulated game environment of DishBrain. Finally, the observations are assumed to be fully observable states, as there is no noise in communicating the states to the chip.

Parameters of interest In this work, we are particularly interested in some parameters of the POMDP critical to decision-making schemes. For the full structure of the generative model, refer to Sec.2.1 in Methods. The parameters of interest are:

- $\mathbb{B}$: The transition dynamics represent the knowledge about state transitions in the system. An informed $\mathbb{B}$ helps predict the future states in the hope of controlling them.
- $\mathbb{CL}$: The counterfactual learning method Isomura et al. [2023], Paul et al. [2024a] (CFL) which depends on the state-action mapping $\mathbb{CL}$ for decision-making.
- Γ : The Risk term in counterfactual learning represents the agent’s confidence. A decrease/increase in Γ can indicate a change in strategy, influencing the learning of $\mathbb{CL}$ mapping used in decision-making.
- $\mathbb{C}$: The prior preference distribution which drives control in the classical active inference agents.

We limited ourselves to the discrete-time, discrete-state POMDP architecture captivated by their explicit structure. Several candidates exist, such as Gaussian mixture models Friston et al. [2018] for continuous problems and ANNs Millidge et al. [2020] for representing the dynamics for high dimensional problems. An interesting observation made while experimenting with active inference models is the influence of tiny changes to the structure and dimensions of the model in performance. The freedom of model structure comes with the cost of fine-tuning details but opens up endless ground for exploration. Algorithms for self-learning the structure of generative models are a promising direction to pursue Friston et al. [2023b].

2.3 AIF: Classical formulation of decision-making in active inference

Active inference is a first-principle approach for modelling behaviour and revolves around a central postulate: every ‘agent’ ensures survival by minimising observations (o) that are high in ‘entropy’ Friston [2010]. This definition calls for a method to evaluate the entropy of an incoming observation (o). The entropy of observation, also known as ‘surprise’ in the active inference literature Friston et al. [2023a], is defined as:

$$S(o) = -\log(P(o)). \quad (2)$$

Recent experimental works Kagan et al. [2022], Strong et al. [2024], Isomura et al. [2023] use the concept of ‘surprise’ in the active inference literature to design feedback signals for embodied agents that demonstrate improved learning with time. In Eq. 2, $P(o)$ represents the probability of a given observation o . It should be noted that an agent is ignorant of the actual likelihood ($P_{true}(o)$) of any observation, as it is not in full control of the environment. The agent is not directly informed of how probable an incoming observation is. Hence, to estimate $P(o)$, the agent must internally encode and use a model to learn and predict the probability of incoming observations. This internal model is called the generative model Friston et al. [2012], Da Costa et al. [2020].

Traditionally, planning and decision-making by active inference agents revolve around minimising the variational free energy of the observations one expects in the *future*. To implement this, we define a policy space comprising sequences of actions in time. The policy space in classical active inference [Sajid et al., 2021] is defined as a collection of policies π_n :

$$\Pi = \{\pi_1, \pi_2, \dots, \pi_N\}, \quad (3)$$

which are themselves sequences of actions indexed in time; that is, $\pi = (u_1, u_2, \dots, u_T)$, where u_t is one of the available actions in U , and T is the agent’s planning horizon. N is the total number of unique policies defined by permutations of available actions u over a time horizon of planning T .

To enable goal-directed behaviour, we must quantify the agent’s preference for sample observations o . The prior preference for observations is usually defined as a categorical distribution over observations,

$$\mathbb{C} = \text{Cat}(o). \quad (4)$$

So, if the value corresponding to an observation in $\mathbb{C}$ is the highest, it is the most preferred observation for the agent. Given these two additional parameters (Π and $\mathbb{C}$), we can define a new quantity called the expected free energy (EFE) of a policy π similar to the definition in [Sajid et al., 2021, Parr and Friston, 2019] as,

$$G(\pi) = \sum_{t=1}^T \underbrace{D_{KL}[Q(o_t|\pi^t) || \mathbb{C}]}_{\text{Risk}} + \underbrace{\mathbb{E}_{Q(s_t|s_{t-1}, \pi^{t-1})} [\mathbb{H}[P(o_t|s_t)]]}_{\text{Expected ambiguity}}. \quad (5)$$

In Eq.(5) above, π^t is the t -th element in π , i.e. the action corresponding to time t for policy π . The term, $Q(o_t|\pi^t)$ represents the most likely observation caused by the policy π at time t . D_{KL} stands for the KL-divergence, which, when minimised, forces the distribution $Q(o_t|\pi^t)$ closer towards $\mathbb{C}$. This term is also called the "Risk," which measures the risk of an action that leads to outcomes that diverge from the agent's preferred states. Thus, actions are selected based on their expected KL divergence — essentially choosing actions that minimize this divergence, thereby minimizing "risk". This represents the goal-directed behaviour of the agent. The KL-divergence between two distributions, P and Q , is defined as:

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}, \quad (6)$$

and $P = Q$ if and only if $D_{KL}(P||Q) = 0$.

In the second term of Eq.5, $\mathbb{H}[P(o_t|s_t)]$ stands for the (Shannon) entropy of $P(o_t|s_t)$ defined as,

$$\mathbb{H}(P(o)) = - \sum_{o \in O} P(o) \log P(o). \quad (7)$$

The second term is also called the 'Expected ambiguity' term. When the expected entropy of $P(o_t|s_t)$ w.r.t the belief $Q(s_t|s_{t-1}, \pi^{t-1})$ is lower, the agent is more confident of the state-observation mapping (i.e., $\mathbb{A}$) in its generative model.

It may already be evident that the above formulation has one fundamental limitation: in the stochastic control problems commonly encountered in practice, the size of possible action spaces, U , and the time horizons of planning, T , make the policy space too large to be computationally tractable. For example, with eight available actions in U and a time horizon of planning $T = 15$, the total number of (definable) policies that need to be considered are $(3.5 * 10^{13})$ 35 trillion. Even for this relatively small-scale example, this policy space is not computationally tractable to simulate agent behaviour (unless additional decision tree search methods [Fountas et al., 2020, Champion et al., 2021a,b] or policy amortisation Fountas et al. [2020], Çatal et al. [2020] are considered or by eliminating implausible policy trajectories using Occam's principle). In this work, we only consider one-step planning as the problem dimensions are intractable for planning with this approach. We refer to the agent using this method for planning as 'AIF-1' abbreviating 'Active inference agent with one-step planning'.

We now describe an improved scheme that uses dynamic programming principles Bellman [1966] to redefine policy space and planning altogether.

2.4 DPEFE: Dynamic programming in expected free energy

In a recent work Paul et al. [2021], an efficient planning algorithm was proposed using dynamic programming principles. The algorithmic formulation generalised for a POMDP setting can be found in Paul et al. [2024b]. We summarise this method below:

For a planning horizon of T (i.e., the agent aims to reach goal state at time T), the EFE of the (last) action for the $T - 1$ th time step can be written as:

$$G(u_{T-1}, o_{T-1}) = D_{KL}[Q(o_T | u_{T-1}, s_{T-1}) || \mathbb{C}]. \quad (8)$$

The term, $G(u_{T-1}|o_{T-1})$ is the expected free energy associated with any action u_{T-1} , given we are in (hidden) state s_{T-1} . This estimate measures how much we believe that the observations at time T will align with our prior preference $\mathbb{C}$.

To estimate $Q(o_T|u_{T-1}, s_{T-1})$, we make use of the prediction about states that can occur at time T , $Q(s_T)$:

$$Q(s_T | u_{T-1}, s_{T-1}) = \mathbb{B}_{u_{T-1}} \cdot Q(s_{T-1}), \quad (9)$$

and given the prediction $Q(s_T)$, we get

$$Q(o_T | u_{T-1}, s_{T-1}) = \mathbb{A} \cdot Q(s_T | u_{T-1}, s_{T-1}) \quad (10)$$

$$= \mathbb{A} \cdot (\mathbb{B}_u^{T-1} \cdot Q(s_{T-1})). \quad (11)$$

Next, using Eq.9, the corresponding action distribution (for action selection) is calculated at time T ,

$$Q(u_{T-1}|o_{T-1}) = \sigma(-G(u_{T-1}|o_{T-1})). \quad (12)$$

Here, we recursively calculate the expected free energy for actions and the corresponding action-distributions for time steps $T-2, T-3, \dots, t=1$ backwards in time,²

$$G(u_t|o_t) = \underbrace{D_{KL}[Q(o_{t+1}|u_t, s_t) || \mathbb{C}]}_{\text{EFE of action at time } t} + \underbrace{\mathbb{E}_{Q(o_{t+1}, u_{t+1}|o_t, u_t)}[G(u_{t+1}|o_{t+1})]}_{\text{EFE of next action at } t+1}. \quad (13)$$

In the equation above, the second term condenses information about all future observations rather than doing a forward tree search in time. To inform $G(u_t|o_t)$, we consider all possible observation-action pairs that can occur in time $t+1$ and use the previously evaluated $G(u_{t+1}|o_{t+1})$. In Eq.13, we evaluate $Q(o_{t+1}, u_{t+1}|o_t, u_t)$ using,

$$Q(s_{t+1}, u_{t+1}|s_t, u_t) = \underbrace{Q(s_{t+1}|s_t, u_t)}_{\mathbb{B}} \cdot \underbrace{Q(u_{t+1}|s_{t+1})}_{\text{Action distribution}}. \quad (14)$$

We then map the distribution $Q(s_{t+1}, u_{t+1}|s_t, u_t)$ to the observation space and evaluate $Q(o_{t+1}, u_{t+1}|o_t, u_t)$ using the likelihood mapping $\mathbb{A}$. In Eq.14, we assume that actions in time are independent of each other, i.e. u_t is independent of u_{t+1} . Even though actions are assumed to be explicitly independent in time, the information (and hence desirability) about actions is also informed backwards from the recursive evaluation of expected free energy.

While evaluating the EFE, G , backwards in time, we used the action distribution in Eq.12. This action distribution can be directly used for action selection. Given an observation o at time t , u_t may be sampled³ from,

$$u_t \sim Q(u_t|o_t = o). \quad (15)$$

We refer to agents using the DPEFE method for decision making as ‘DP-T’, abbreviating ‘Active inference agent using dynamic programming with planning horizon T ’.

2.5 CFL: Counterfactual learning

A different approach to decision-making does not rely on planning but on counterfactual learning (CFL). In the counterfactual learning method, the agent learns a state-action mapping $\mathbb{CL}$ instead of evaluating EFE directly⁴. This state-action mapping is learned using a ‘Risk’ parameter $\Gamma(t)$ using the update equation as given in Isomura and Friston [2020] as:

$$\mathbb{CL} \leftarrow \mathbb{CL} + t \langle (1 - 2\Gamma(t)) \langle u_t \otimes s_{t-1} \rangle \rangle. \quad (16)$$

Here, $\langle \cdot \rangle$ refers to the average over time, and $\otimes$ is the Kronecker-product operator. Given the state-action mapping $\mathbb{CL}$, agent samples actions from the distribution,

$$P(u|s_{t-1})_{CL} = \sigma(\ln \mathbb{CL} \cdot s_{t-1}). \quad (17)$$

The free parameter in our model is the number of past instances (of state-action pairs) the agent stores in memory use in every time-step to learn $\mathbb{CL}$ in Eq.16. In this paper, ‘CFL-T’ represents the active inference agent with a memory horizon of T .

The functional form of $\Gamma(t)$ used in the simulations of this work is:

$$\Gamma(t)_{prior} = 0.55 \quad (18)$$

A value of Γ greater than 0.55 corresponds to “higher risk” in the CFL method. An initial value greater than 0.5 is necessary to enable learning Isomura and Friston [2020].

²For times other than $T-1$, the first term in Eq.13 does not contribute to solving the particular instance if $\mathbb{C}$ only accommodates preference to a (time-independent) goal-state. However, for a temporally informed $\mathbb{C}$, i.e. with a separate preference for reward maximisation at each time step, this term will meaningfully influence action selection.

³Precision of action-selection may be controlled by introducing a positive constant inside the softmax function $\sigma(\cdot)$ in Eq.12. The higher the constant, the higher the chance of selecting the action with less EFE.

⁴For the exact form of the generative model and free energy, refer to Isomura and Friston [2020]

For updating Γ , we use the equation,

$$\Gamma(t) \leftarrow \Gamma(t) - \frac{1}{T_{goal} - t}. \quad (19)$$

Here, T_{goal} is when the agent receives a positive environmental reward. So, the sooner the agent reaches the ‘goal-state’, the quicker the $\Gamma(t)$, i.e., risk converges to zero. All the update rules this paper defines can be derived from the postulate that the agent tries to minimise the (variational) free energy w.r.t the generative model Isomura et al. [2023], Paul et al. [2024b]. It was observed in Paul et al. [2024a] and Khajehnejad et al. [2024] that this variation of the CFL method is well suited for environments that require spontaneous decision-making and is data-efficient (Environments like the Cart Pole (OpenAIGym) Barto et al. [1983], game of Pong Khajehnejad et al. [2024]).

The algorithms implemented in this paper for decision-making (DPEFE Paul et al. [2024b] and CFL Paul et al. [2024b]) are low-cost computational methods against classical active inference algorithms. The improvement in computational complexity of these models enabled simulations in the state space consisting of more than two thousand states and has the potential to scale to more demanding environments.

2.6 Explainability: Normalised total entropy

To examine some parameters, we define the total entropy (TE) of a parameter as,

$$\text{TE}(Z) = \sum_{X \in Z} \mathbb{H}(X) = - \sum_{X \in Z} \sum_{x \in X} p(x) \log p(x). \quad (20)$$

In Eq.20, Z is a parameter like the $\mathbb{CL}$ mapping in the counterfactual learning method, and $\mathbb{H}$, the Shannon entropy. X is a probability distribution collection comprising the parameter Z . For example, the $\mathbb{CL}$ mapping is made up of probability distributions $P(u|s_t) \forall s \in S$, representing how probable an action u is, given a state s at time t . For visualising the evolution of total entropy with time, we scale the vector storing total entropies in each time step to values between zero and one, i.e. normalise⁵ it. Normalisation is advantageous to us here because it emphasises the underlying trends rather than the absolute values. Scaling also enables comparison between groups. We call this measure the normalised total entropy (NTE).

3 Results

3.1 Counterfactual learning method with memory demonstrates significant improvement in learning

Firstly, we simulate active inference agents using the counterfactual learning algorithm Isomura et al. [2023] for decision-making. We use a memory-augmented version of the algorithm developed in Paul et al. [2024a], in which the agent uses the last T time steps for learning the $\mathbb{CL}$ mapping, referred to as ‘CFL-T’ in the paper. For more details, refer to Sec.2.5 in Methods.

To standardise the simulation results against the experimental results, we time-stamp every trial, matching a total trial length of twenty minutes as in the DishBrain experiments. In the biological experiments, an average of 69.04 ± 7.95 game episodes comprised a trial of 20 minutes Kagan et al. [2022], Khajehnejad et al. [2024]. In our simulations, we conduct trials with 70 episodes each and add time labels to match a total trial length of 20 minutes. To ensure reproducibility and minimise the impact of hyperparameter selection, we conduct 100 trials with different random seeds. Given the assigned time stamps, we can compare the performance of in vitro agents to our in silico agents.

From Fig.2, we observe improvement in the performance of ‘CFL-T’ agents with higher memory horizon (T) across all game metrics. Considering only lower memory horizons (top row), the CFL-4 agent outperforms the ‘biological agents’ (HCC and MCC) across all the performance matrices. The memory horizon of 4 and above essentially implies a higher ability to use the information from the past to improve performance. However, a memory horizon of 4 may not be biologically plausible, given that the presence of long-term memory in the DishBrain system has not yet been demonstrated. Notably, CFL agents with the lowest memory horizons (1-2) perform at par with the MCC and HCC groups, and these observations motivate us to study the role of memory in systems like the DishBrain and inspire future experiments dependent on memory.

In Fig.2 (bottom row), we observe an improvement in performance with a higher memory horizon. Long memory horizons, for example, 16 and 32, are biologically implausible for systems like DishBrain. However, these results stress the relevance of further studying the role of memory in such embodied systems. In Sec. 3.2, we study the performance

⁵To normalise a vector V , we use the sklearn machine toolkit, specifically: `sklearn.preprocessing.normalize(V, norm='max')`.

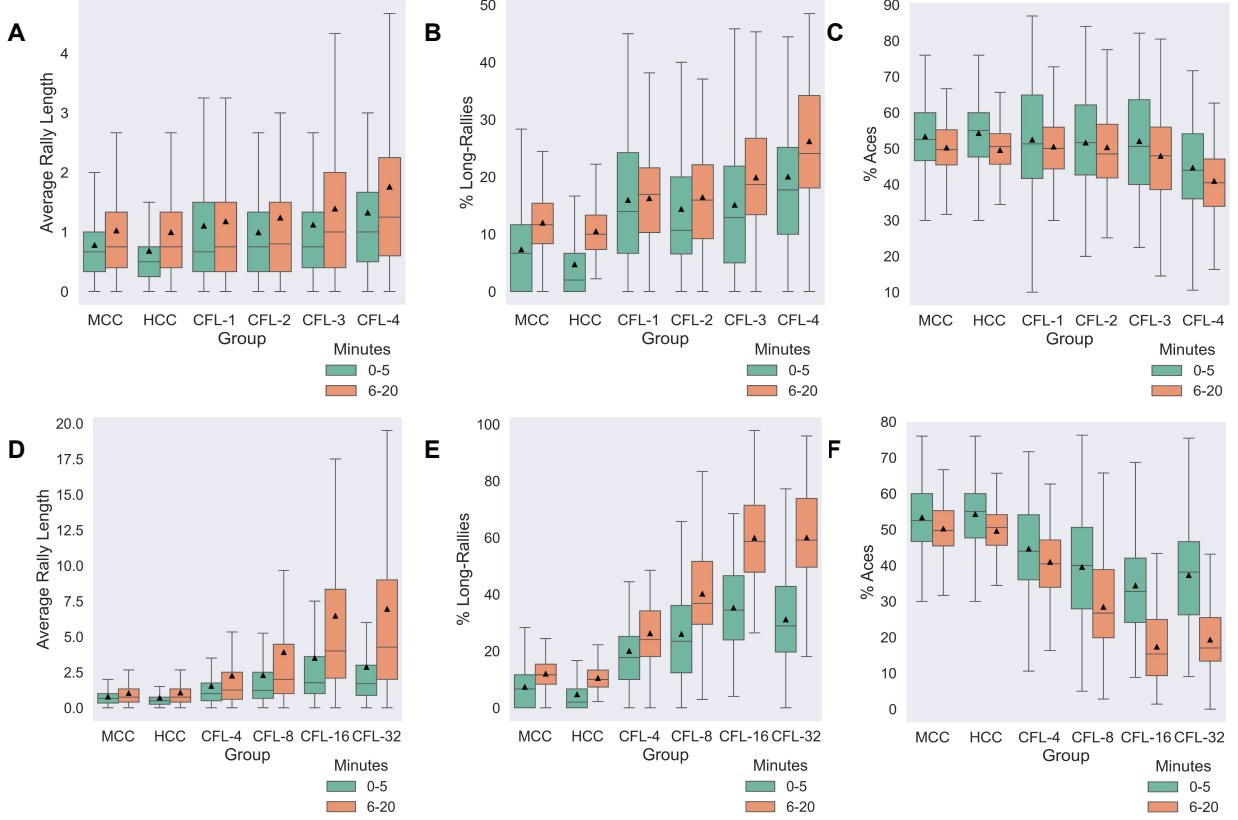


Figure 2: Memory horizon influences performance in CFL agents compared with MCC and HCC groups. Changes in agent performance are assessed by comparing the first 5 minutes (green) to the last 15 minutes (orange) of the trials. The average number of **A, D**: hits per rally, **B, E**: percentage of aces (balls missed after the initial serve), and **C, F**: percentage of long rallies (≥ 3 consecutive hits) are plotted over a 20-minute real-time equivalent for CFL agents and biological MCC and HCC cultures. **Top row (A, B, C)**: CFL agents with shorter memory horizons exhibit a linear improvement in performance across all game metrics, with CFL-4 outperforming MCC and HCC groups. However, memory horizons of 4 or higher may not be biologically plausible, as long-term memory has not yet been demonstrated in DishBrain. **Bottom row (D, E, F)**: Agents with longer memory horizons (e.g., 16 and 32) show continued performance gains, although such horizons are implausible for DishBrain-like systems. These findings highlight the role of memory in decision-making under active inference and suggest avenues for further experimental investigations in embodied biological systems.

of different decision-making algorithms in active inference and interpret their performance. We use linear regression to further explore agents’ performance in continuous time (See Fig.B.2). Next, we perform a comparison across other popular active inference algorithms.

3.2 Comparison across other popular active inference algorithms

In this section, we compare the classical decision-making scheme in active inference (AIF-1) Sajid et al. [2022], Da Costa et al. [2020] and dynamic programming in expected free energy (DPEFE) Paul et al. [2024b], Friston et al. [2023c]. For more details, refer to Sec.2.3 and Sec.2.4 in Methods.

Fig. 3 summarises the simulation results with the above-mentioned agents. For these figures, we select agents with performance comparable to the ‘MCC’ and ‘HCC’ groups. From Fig. 3 we observe the following:

- From Fig. 3 (top row), we observe that all ‘DP-T’ agents improve similarly to the ‘HCC’ group. A higher planning horizon does not seem to help significantly with performance. Also, over a planning horizon of 10, the performance declines. This effect can be attributed to over-planning.

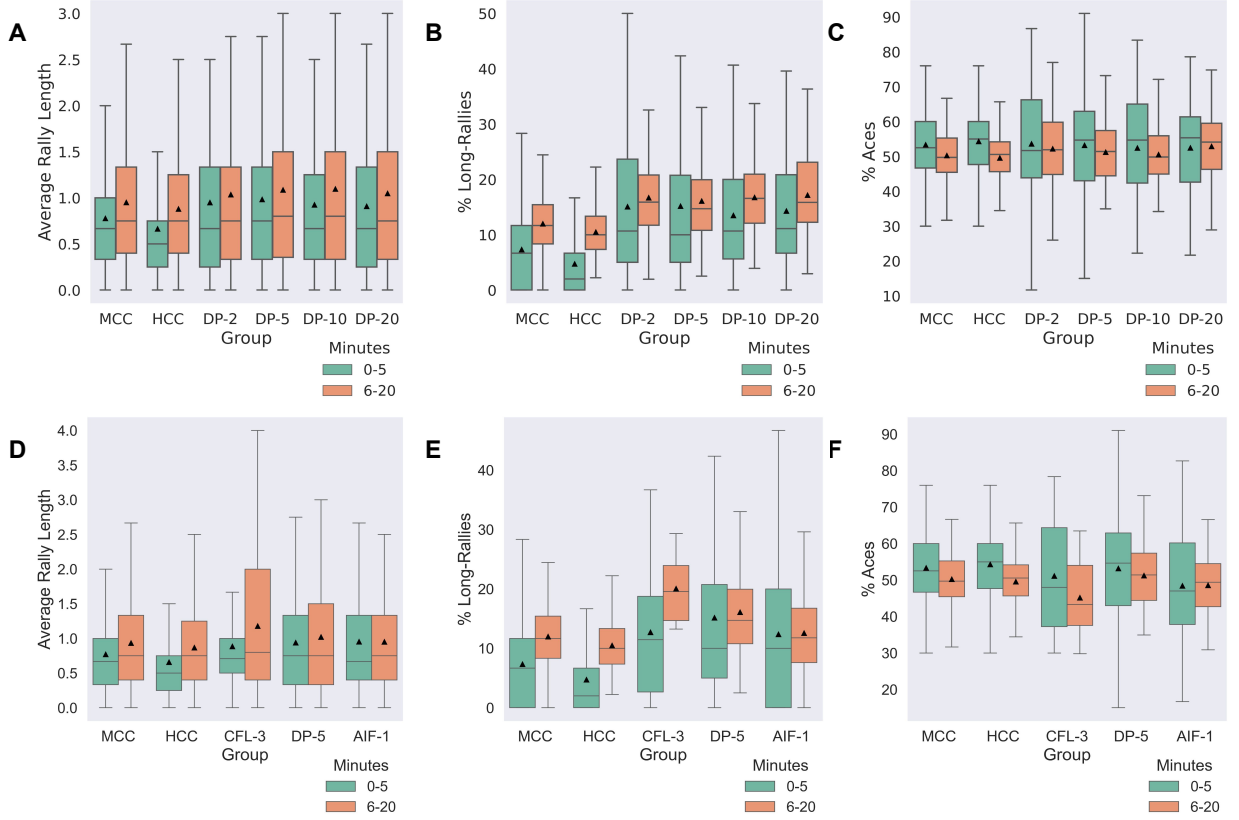


Figure 3: Performance of in-silico active inference agents compared to in-vitro MCC and HCC groups in a simulated Pong environment. The game performance is evaluated across three key metrics: **A, D:** Average rally length (higher is better), **B, E:** Percentage of long rallies (higher is better), and **C, F:** Percentage of aces (lower is better). **Top row (A, B, C):** Active inference agents with different planning horizons (‘AIF-1’ and ‘DP-T’) show improvements comparable to the HCC group. However, a planning horizon beyond 10 leads to a decline in performance, likely due to over-planning. Additionally, DPEFE agents perform similarly across the task, suggesting that increased planning depth does not significantly enhance performance in Pong. **Bottom row (D, E, F):** The ‘CFL-3’ agent outperforms all other groups across all game metrics, reinforcing the idea that memory-based decision-making is more effective for in-game environments like Pong than planning-based approaches. These results highlight the trade-offs between memory and planning in active inference-based control.

- Similarly, from Fig.3 (top row), all DPEFE agents perform similarly in the pong game task. This suggests that a higher planning horizon is not helpful in the pong game environment. This observation suggests that memory-based decision-making is more useful for in-game environments like Pong than planning algorithms like DPEFE.
- From Fig.3 (bottom row), ‘CFL-3’ group outperforms other groups in all measured game metrics (‘Average rally length’, ‘% of Long Rallies’, and ‘% of Aces’).
- From Fig.3 (bottom row), we observe that the ‘AIF-1’ agent demonstrates a relative improvement comparable with the control ‘CTL’ group. Evidently, with one-step planning, the agent seems to make no difference in performance over time.

See Fig. B.1 for a different representation of the same data. In Fig. B.1 (C), we compare the relative improvement of CFL agents with different memory horizons. We see that ‘CFL-4’ outperforms every other group regarding relative improvement.

In the next section, we examine the evolution of model parameters and draw insights from them. We leverage the model parameters to understand the basis of performance on a deeper level.

3.3 Model parameters reveal explainable insights into the performance of the active inference agents

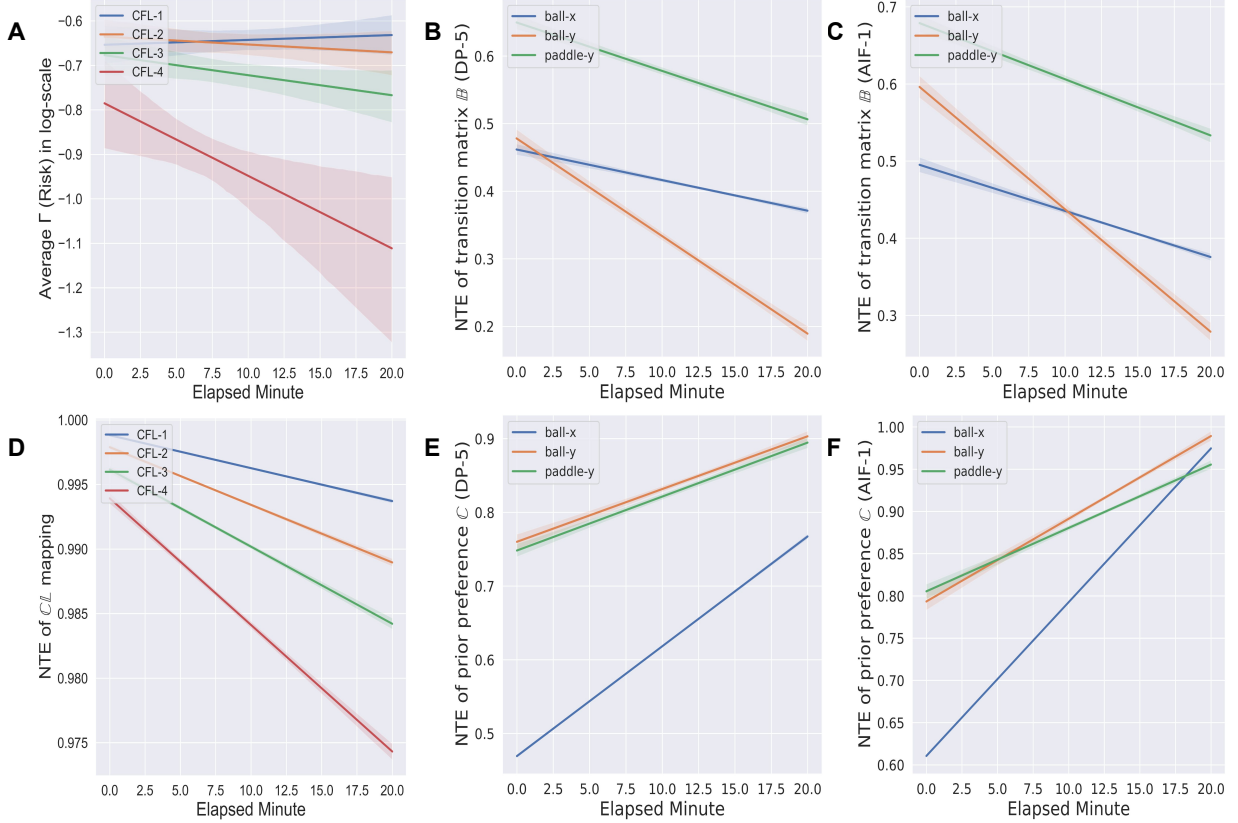


Figure 4: Evolution of key parameters in CFL, DP-5, and AIF-1 agents. **A:** Evolution of the average risk term (Γ_t) in CFL agents (log scale). A significant drop in risk is observed only for the ‘CFL-4’ agent, aligning with its superior performance. **B, C:** Evolution of the normalised total entropy (NTE) of the transition dynamics for **B:** DP-5 and **C:** AIF-1 agents. The decreasing entropy in both cases indicates that agents are progressively learning about the environment. **D, E, F:** Evolution of the NTE for key model parameters: the CL vector (**D**), and prior preferences (**E, F**) in DP-5 and AIF-1 agents, respectively. **D:** The CL vector’s entropy decreases significantly for higher memory horizons, highlighting goal-directed learning in CFL agents. **E, F:** In both DP-5 and AIF-1 agents, the entropy of prior preference distributions increases over time, suggesting that no specific ball or paddle position is inherently preferred, reinforcing that the primary objective is defending the ball rather than favouring particular game states. These findings underscore differences in how memory-based (CFL) and planning-based (DP-5, AIF-1) agents learn and adapt in the game environment.

Since all the parameters in our models are non-approximate, we can examine how they evolve to draw insights into the performance and learning of the agents. We examine the parameters in time frames similar to the biological experiment.

In Fig. 4 (A), we look at the average Γ_t (Risk) in CFL agents on a log scale. We see a significant drop in the risk parameter over time, only for the ‘CFL-4’ agent. This observation qualitatively matches the significant increase in performance of the ‘CFL-4’ agent as observed in Fig. 2.

To study the evolution of parameters that encode probability distributions (such as $\mathbb{B}$, $\mathbb{C}$), we propose a measure called the ‘Normalised total entropy (NTE)’ in this paper. For more details, see Sec.2.6. NTE represents how informed a probability distribution is; hence, a lower NTE corresponds to a more informed distribution relative to an uninformed prior with the highest entropy. In Fig. 4 (D), we see that the NTE of the $\mathbb{C}$ L vector decreases with time and has the highest decrease for higher memory horizons. This is evidence for the emergence of goal-directed learning in the CFL agents and also results in improved performance over time, which is observed in Fig.2. We also observe the significant decrease in the NTE for the ‘CFL-4’ agent in Fig.4 (D) matching the higher performance of ‘CFL-4’ in Fig.2.

Additionally, the dynamic programming agent’s model parameters reveal the method’s limitation in this environment. In Fig. 4 (second column), we examine the parameters of the DP-5 agent. There is no particular reason for choosing this group over other DP-T groups. In Fig. 4 (B), we look at the normalised total entropy of the transition dynamics $\mathbb{B}$. The entropy corresponding to all state-observation modalities decreases, indicating that the agent is learning about the environment. A decreased entropy represents the probability distributions becoming more informed than a uniform prior with maximum entropy. Similarly, in Fig. 4 (E), we see that the entropy of the prior preference distribution $\mathbb{C}$ increases for all state-observation modalities. This observation is counter-intuitive as we would expect learning of some sort, but a possible explanation is that learning a prior preference is not possible in this game since no particular ball or paddle position is preferred. Only defending the ball in the game gives positive rewards, and no particular position of the ball and paddle is advantageous. We interpret it as the agent is not learning a particular preference about any of the observations, as defending the ball is the goal, and a particular position of the ball and paddle needs to be favoured.

Similar to the DP-5 agent, in Fig.4 (last column), we examine the model parameters of the ‘AIF-1’ agent and observe a similar trend. From Fig.4 (C), we see that the entropy of the transition dynamics of all modalities decreases, which provides evidence that the agent is learning more about the environment. Interestingly, the entropy of the prior preference distribution increases, as seen in Fig.4(B).

We leave the detailed analysis of parameters and testing different structures of the generative models to future work. The insights from this work will enable analysis of similar future experiments, opening a paradigm of probing into the basis of intelligence.

4 Discussion

We successfully demonstrate the efficacy of the active inference framework in modelling purposeful decision-making within a synthetic biological intelligence context, marking a significant step in understanding such systems. Our simulations reveal that agents employing memory-based learning and predictive planning can efficiently navigate and adapt within dynamic environments. The active inference approach provides a biologically plausible and inherently interpretable model for how systems, akin to biological neuronal networks, can achieve continuous learning and real-time responsiveness. By unifying perception, action, and learning under a single probabilistic calculus, active inference emerges as a compelling alternative to traditional machine learning paradigms, particularly for developing biologically inspired AI.

The advantages of this approach are underscored when considering the remarkable sample efficiency of biological systems, a trait many artificial systems, including deep reinforcement learning (RL) agents, struggle to emulate Khajehnejad et al. [2024]. Our adoption of active inference aims to capture not only this efficiency but also the crucial elements of biological plausibility and interpretability, aspects where opaque, extensively-tuned RL models often fall short Friston [2010, 2012]. This makes active inference particularly well-suited for modelling systems like DishBrain, where the interplay of external stimuli and emergent behaviour necessitates a probabilistic framework adept at managing uncertainty and facilitating structured learning.

The analysis of key model parameters, such as the risk term (Γ) and the normalized total entropy (NTE) of the $\mathbb{CL}$ vector, offers explainable insights into the agents’ learning dynamics. For instance, the observed reduction in risk and NTE in high-performing agents signifies increasing confidence and refinement in their decision-making strategies over time Paul et al. [2024a], supporting the emergence of goal-directed behaviour. These findings are pivotal for advancing safe and transparent AI systems Albarracín et al. [2023], as they allow us to trace how uncertainty is minimised through adaptive exploration, thereby bridging experimental neuroscience with theoretical models of intelligence.

A critical consideration for future work is the biological instantiation of memory mechanisms. While our simulations show enhanced performance in CFL agents with extended memory, current experimental evidence from systems like DishBrain indicates rapid network reorganisation and population-wide dynamics rather than explicit memory recall Kagan et al. [2022], Habibollahi et al. [2023]. This highlights an important avenue for investigation: determining whether sophisticated memory, as modelled, is essential for biological learning, or if rapid adaptive restructuring alone underpins the observed behaviours.

Future research should therefore focus on dissecting the roles of different memory types and adaptive dynamics in both synthetic and biological agents. Learning more about these distinctions will be instrumental in refining our models and could pave the way for innovative hybrid systems that synergise the adaptive strengths of biological intelligence with the explanatory power of computational frameworks like active inference. Ultimately, this work contributes to a more integrated understanding of purposeful behaviour, pushing forward the development of AI that is not only intelligent but also transparent and grounded in biological principles.

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6 Software note

All the code for active inference agents, pong-game environment, and visualisation used are custom written in Python 3.9.15 and is available in this project repository: https://github.com/aswinpaul/pong_ai_2023.

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A An overview of the DishBrain experiment and results

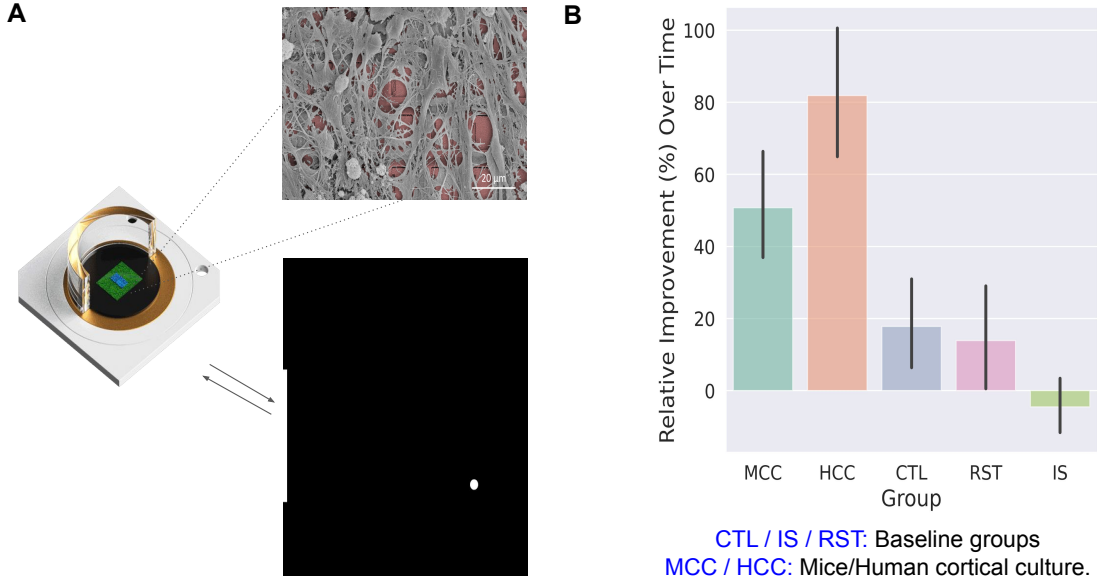


Figure A.1: **A:** A high-level schematic of the ‘DishBrain’ experiment. Live neuron populations are cultured on a silicon chip and embodied in a simulated game environment of ‘Pong’, where they receive information about game states and feedback signals to control the paddle through electrophysiological stimulation and recording. **B:** Relative improvement of groups in one of the game metrics (average rally length). We observe that both the mouse cortical culture (‘MCC’) and human cortical culture (‘HCC’) groups demonstrate improvement in performance with time. The performance is compared across two blocks of time in the experiment: the first five minutes and the next fifteen minutes of the twenty-minute trial. The ‘HCC’ group demonstrated higher relative performance improvement than other groups. The baseline groups are control (‘CTL’), in-silico (‘IS’), and rest (‘RST’). Figures adapted from Kagan et al. [2022].

The ‘DishBrain’ system (Fig. A.1 (A)) contains active neuron populations coupled to a simulated game environment of ‘Pong’. In this setup, the neurons control paddle movements with the goal of intercepting the ball. With no opponent player, the ball bounces off the back wall, returning towards the paddle after each successful hit. When the ball is missed, the game restarts with the ball at a random location. The game performance is evaluated across three key metrics: average rally length (longer rallies indicating better performance), percentage of long rallies (the proportion of rallies lasting more than three hits, higher the better), and percentage of aces (instances where the ball is missed immediately, with fewer aces being better). The ball’s x and y-axis location is encoded to the neurons via electrical stimulation, combining place-coded and rate-coded signals. Through game-outcome dependent feedback, learning and improvement in the performance of ‘MCC’ and ‘HCC’⁶ is reported as measurable improvements, particularly in average rally length, as demonstrated in Fig. A.1 (B).

DishBrain is an ideal platform for studying the emergence of purposeful behaviour, as it demonstrates the ability to learn and adapt to its environment de novo without any prior exposure. To theoretically analyse this phenomenon, we turn to active inference, a first principle approach to modelling behaviour that emerged in neuroscience and was initially proposed as a unified theory of brain function Friston [2010, 2012]. Active inference posits that behaviour is driven by an agent’s beliefs about its environment rather than being a separate process, as seen in frameworks like reinforcement learning Friston et al. [2009], Neftci and Averbeck [2019]. This approach allows us to model the dynamics of learning and decision-making in a biologically plausible manner, offering a deeper understanding of how systems like DishBrain exhibit adaptive behaviour.

⁶‘MCC’ and ‘HCC’: Mouse and human cortical cells

B Additional figures

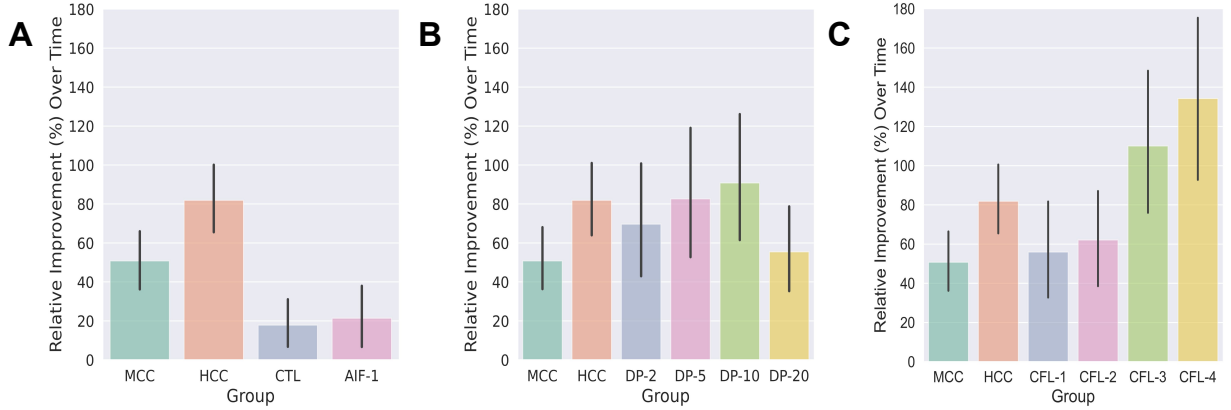


Figure B.1: **Relative improvement in performance (Average rally length) of in-silico active inference agents compared to in-vitro groups of interest in the simulated game environment of Pong.** **A:** Performance of the ‘AIF-1’ agent compared to other groups across key game metrics. The AIF-1 agent demonstrates a relative improvement over time, but its planning-based approach does not yield significant advantages in the Pong environment, suggesting that memory-based strategies may be more beneficial. **B:** Performance of ‘DP-T’ agents against significant groups. While DP-T agents exhibit initial learning, performance gains plateau with increased planning horizons, and over-planning (beyond a horizon of 10) results in diminished effectiveness, likely due to over-planning. **C:** Performance of ‘CFL-T’ agents compared to significant groups. CFL agents with higher memory horizons consistently outperform other groups across all metrics, reinforcing the role of memory in adaptive decision-making. These findings highlight the trade-offs between planning-based and memory-driven decision-making approaches, with memory-based strategies proving more effective in real-time interactive tasks like Pong.

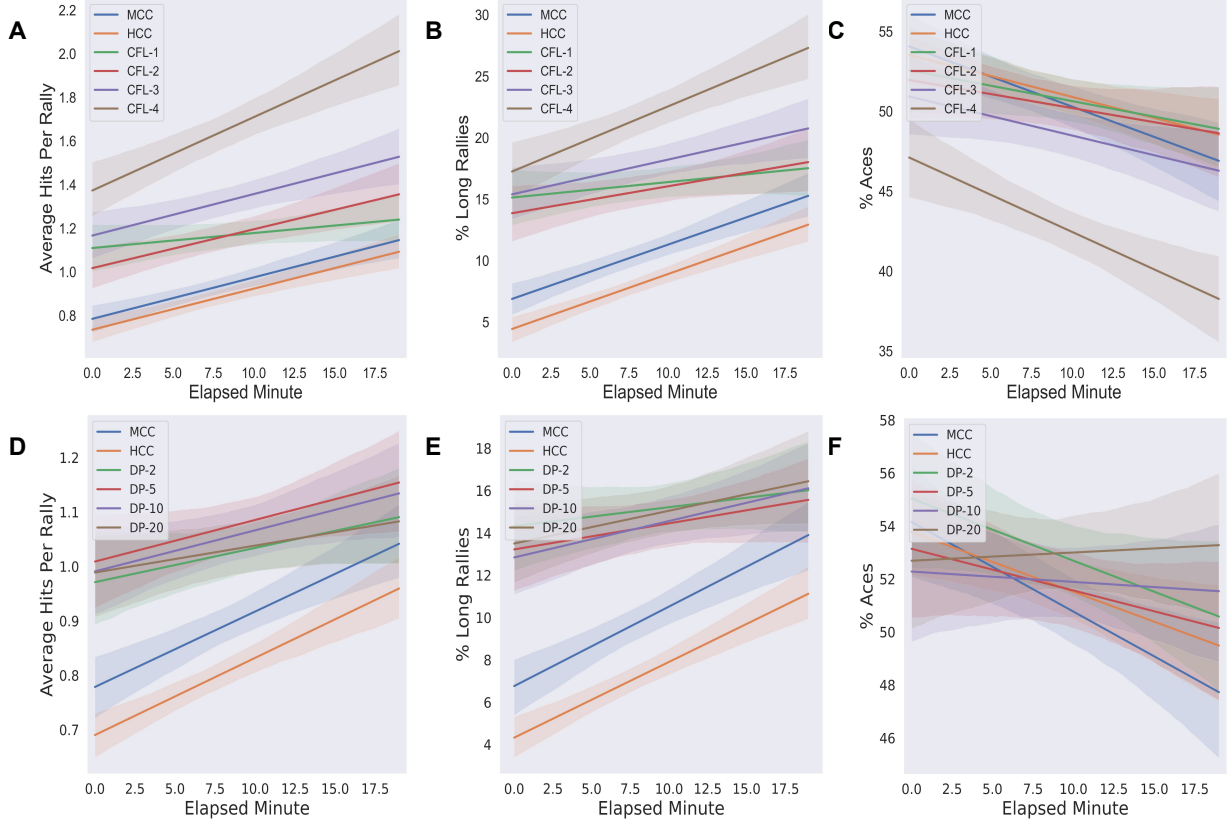


Figure B.2: **Continuous-time analysis of memory horizon effects on CFL agent performance.** Changes in agent performance are tracked over a 20-minute real-time equivalent using linear regression. The average number of **A, D**: hits per rally, **B, E**: percentage of aces (balls missed after the initial serve), and **C, F**: percentage of long rallies (≥ 3 consecutive hits) are analyzed for CFL agents and biological MCC and HCC cultures. **Top row (A, B, C)**: CFL agents with shorter memory horizons exhibit a steady, linear improvement across all game metrics, with CFL-4 surpassing MCC and HCC groups. However, memory horizons of 4 or greater may not be biologically plausible, as DishBrain’s long-term memory capabilities remain to be studied. **Bottom row (D, E, F)**: Agents with longer memory horizons (e.g., 16 and 32) continue to show performance gains, though such horizons are implausible for DishBrain-like systems. These results reinforce the role of memory in decision-making under active inference and provide a dynamic, time-continuous perspective on learning in embodied systems.

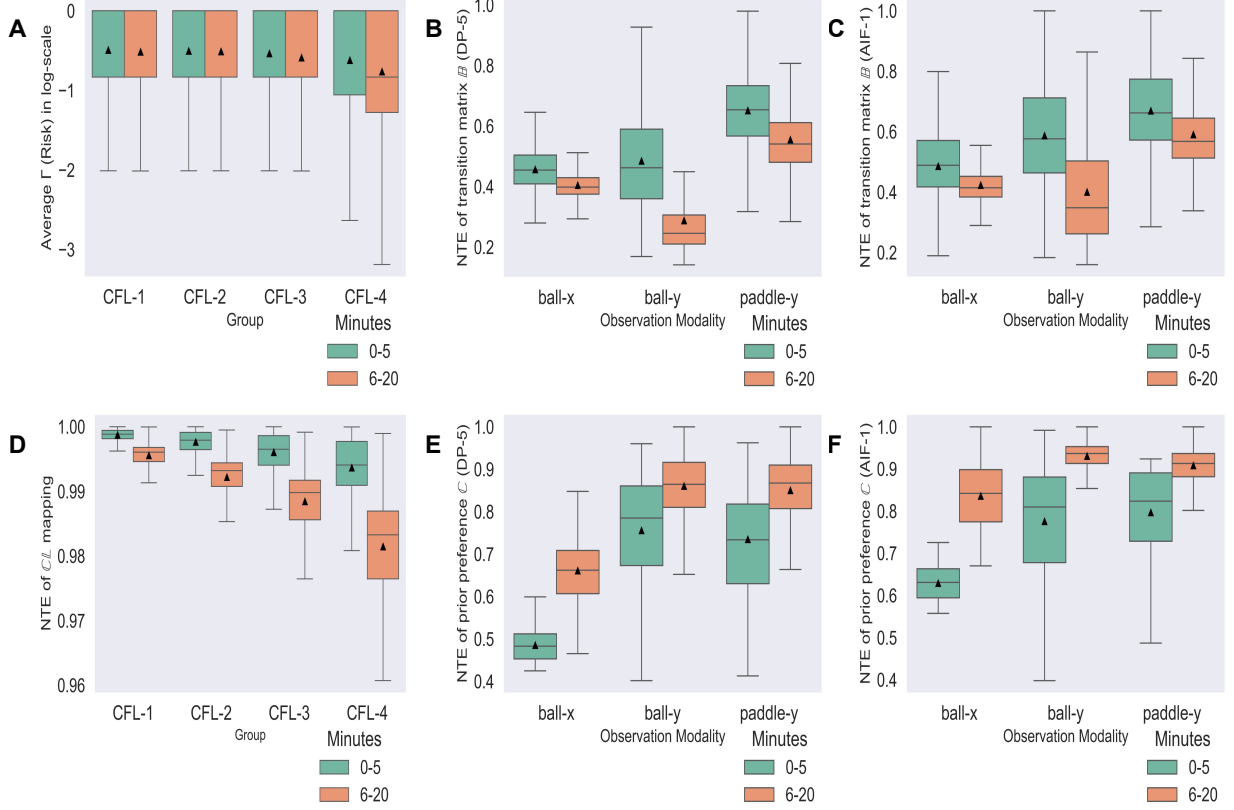


Figure B.3: **Box plot comparison of in-silico active inference agents and in-vitro MCC and HCC groups in a simulated Pong environment.** Game performance is summarized using box plots across three key metrics: **A, D:** Average rally length (higher is better), **B, E:** Percentage of long rallies (higher is better), and **C, F:** Percentage of aces (lower is better). **Top row (A, B, C):** Active inference agents with varying planning horizons ('AIF-1' and 'DP-T') exhibit performance improvements comparable to the HCC group. However, planning horizons exceeding 10 lead to a decline in performance, likely due to over-planning. Additionally, DPEFE agents show consistent performance across all metrics, suggesting that deeper planning does not yield significant benefits in this task. **Bottom row (D, E, F):** The 'CFL-3' agent achieves the highest performance across all metrics, further emphasizing that memory-based decision-making is more effective than planning-based approaches in dynamic environments like Pong. Box plots provide a clearer visualization of variability and consistency across trials, reinforcing the observed trade-offs between memory and planning in active inference-based control.